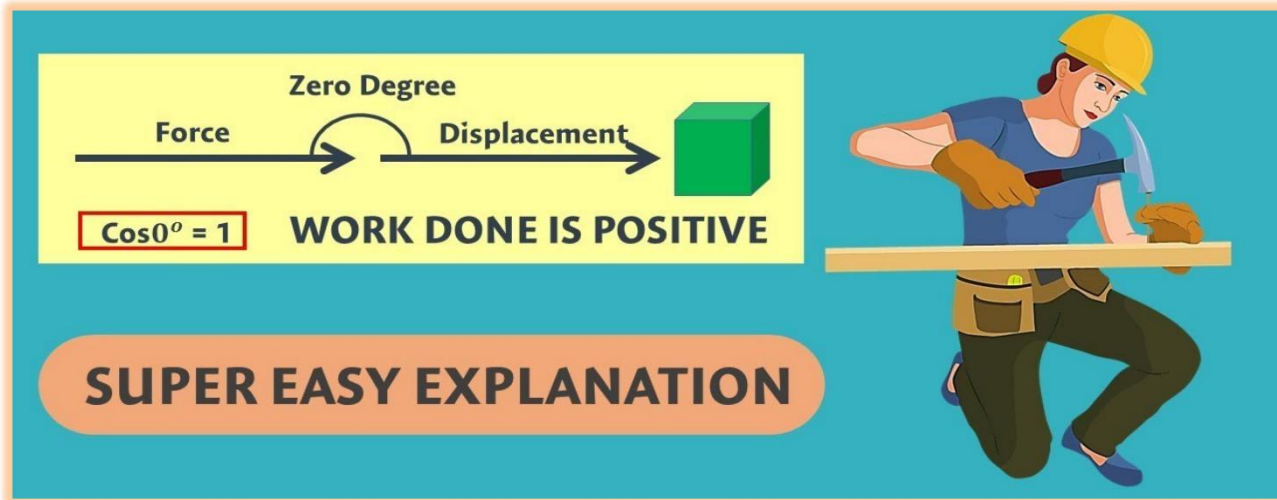




5.1 Work

5.2 Energy and

Conservation of Energy





Learning Outcomes

5.1 Work

At the end of this chapter, students should be able to:

- (a) **Define** work done by a constant force.
- (b) **Explain** the physical meaning of the dot product

$$W = \vec{F} \bullet \vec{s}$$

- (c) **Use** the equation for work done by a constant force,

$$W = Fs \cos \theta$$

- (d) **Determine** work done from F - s graph.



5.2 Energy and Conservation of Energy

At the end of this chapter, students should be able to:

(a) **Define**

- i) kinetic energy
- ii) gravitational potential energy
- iii) elastic potential energy

(b) **Use**

- i) kinetic energy $K = \frac{1}{2}mv^2$
- ii) gravitational potential energy $U_g = mgh$
- iii) elastic potential energy $U_s = \frac{1}{2}kx^2$

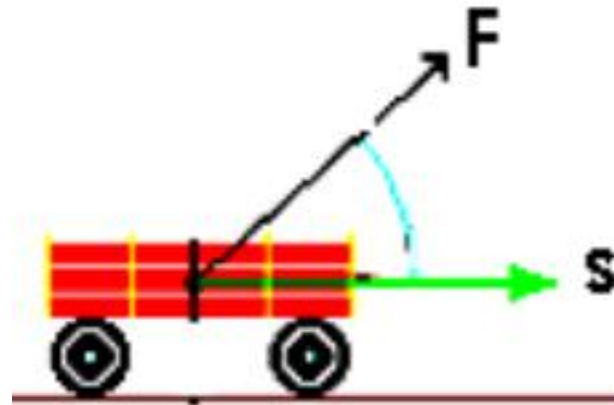
(c) **State** the principle of conservation of energy.

(d) **Apply** the principle of conservation of energy.

What is WORK?

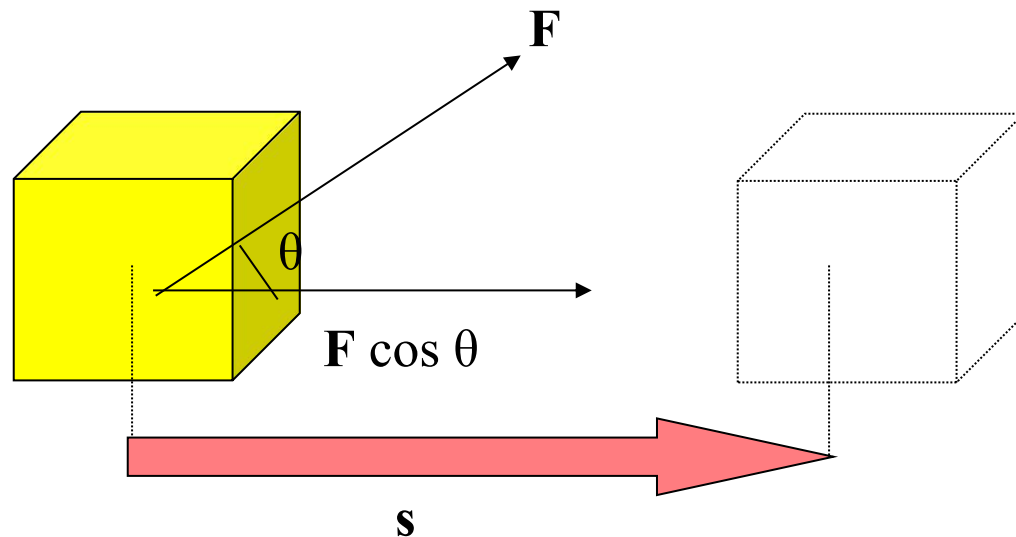
Work is the dot (scalar) product of the force and displacement.

$$W = \vec{F} \bullet \vec{s}$$



Work done by a constant force

If an object undergoes a displacement of s , the work done by the force F is given by



$$W = (F \cos \theta)s$$

where

F = magnitude of force

s = displacement of the body

θ = angle between F and s

- Definition of Work, W done by a constant force acting on an object is **equal** to the **product of the magnitudes of the displacement** and the component of the **force parallel to that displacement**.
- Work is a **scalar** quantity.
- The SI units of work are **joules (J)**.
 - * 1 Joule is the work done when a **force of 1 N moves** through a **distance of 1 m** in the direction of the force.

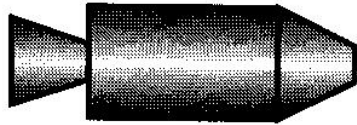
Caution :

- **Work done** on an object is **zero** when **$F = 0$** or **$s = 0$** and **$\theta = 90^\circ$** .

$$F = 0$$

work done = 0
on spacecraft

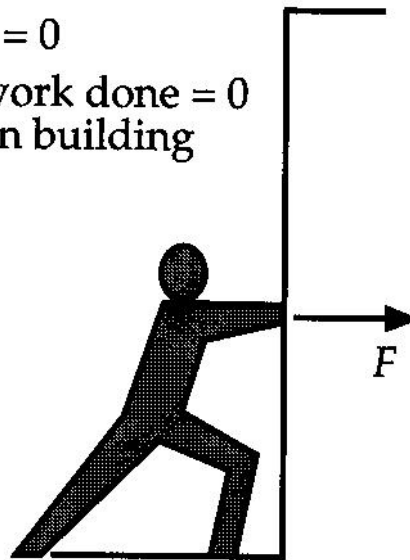
constant
velocity →



Spacecraft
travelling with
constant velocity

$$s = 0$$

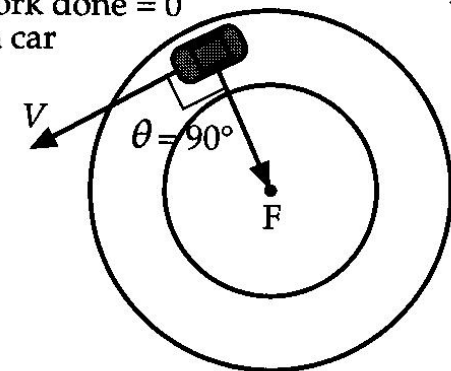
work done = 0
on building



Force exerted on
a building

$$\theta = 90^\circ$$

work done = 0
on car



Centripetal force acting
on the car moving
with constant speed in
a circle

Sign for Work Done, W

Work does not have direction. However, it can have positive, zero or negative value depending on the angle θ between F and s .

If $0^\circ < \theta < 90^\circ$ then
 $\cos \theta > 0$ (positive value)

therefore $W > 0$ (positive)

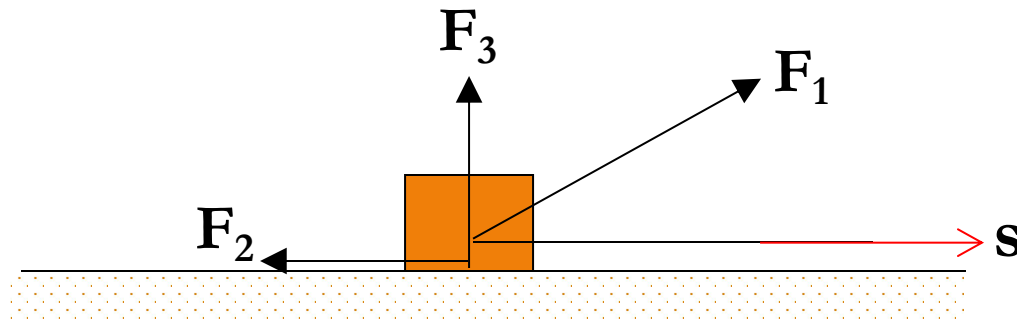
$\Rightarrow$ work done **on the system**
(by the external force) where
energy is transferred to the
system.

If $90^\circ < \theta < 180^\circ$ then
 $\cos \theta < 0$ (negative value)

therefore $W < 0$ (negative)

$\Rightarrow$ work done **by the system**
where energy is transferred
from the system.

- ❖ If more than 1 force acts on an object, the total work done (or net work done) is the **sum** of the work done by each force separately.



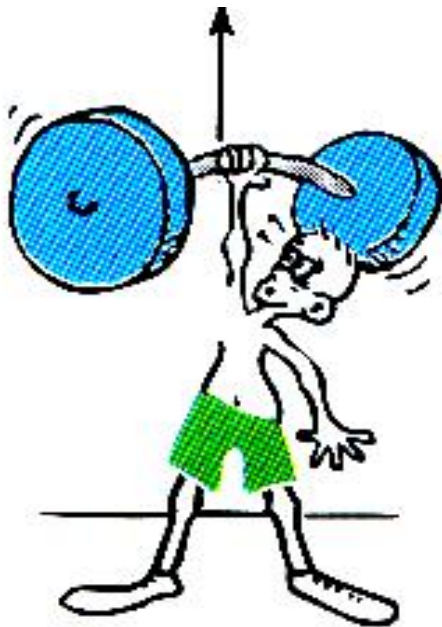
$$W_{Total} = W_{F1} + W_{F2} + W_{F3} + \dots$$

or

$$W_{Total} = F_{net}(s)$$

Conceptual Example

A weightlifter raises a barbell in such a way that it moves up at a constant speed. While it is moving, what is the total work done on the barbell ?



Answer :

Zero.

Since it moves with constant speed, acceleration is zero $\rightarrow$ net force acting on it is zero.

From: $W_{\text{net}} = F_{\text{net}} (s)$

Thus the total work done is zero.

Work done by a variable force

Force-distance graph

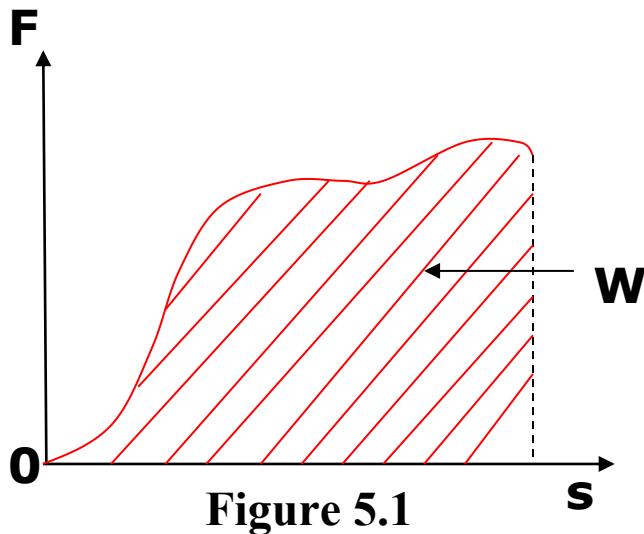


Figure 5.1 shows a force, F whose magnitude changes with the displacement, s .

$$dW = \vec{F} \bullet d\vec{s}$$

$$W = \int_b dW$$

$$W = \int_a \vec{F} \bullet d\vec{s}$$

If the force F varies with displacement s ,

Work done = area under the $F - s$ graph

5.2 Energy and Conservation of Energy

Energy

- Doing work requires energy.
- A body which is capable of doing work is said to possess energy.
- The amount of energy that a body has is equal to the amount of work that it can do.

Energy is the ability to do work.

- It is a scalar quantity
- S.I unit for energy is **joule (J)**
- 2 types of mechanical energy:
 - (1) Potential Energy, U
 - (2) Kinetic Energy, K

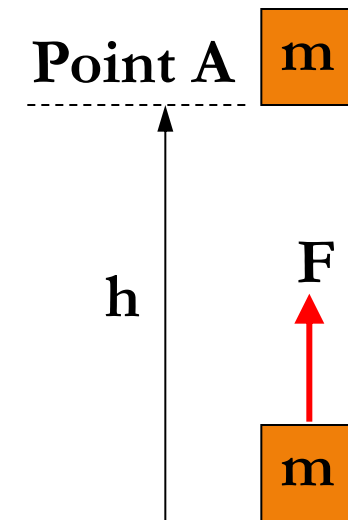
Notes:

Doing work involves a transfer of energy from one form to another form.

Work done = Energy transferred

Example

Lifting up a box from floor to point A. Work done change to potential energy at point A.



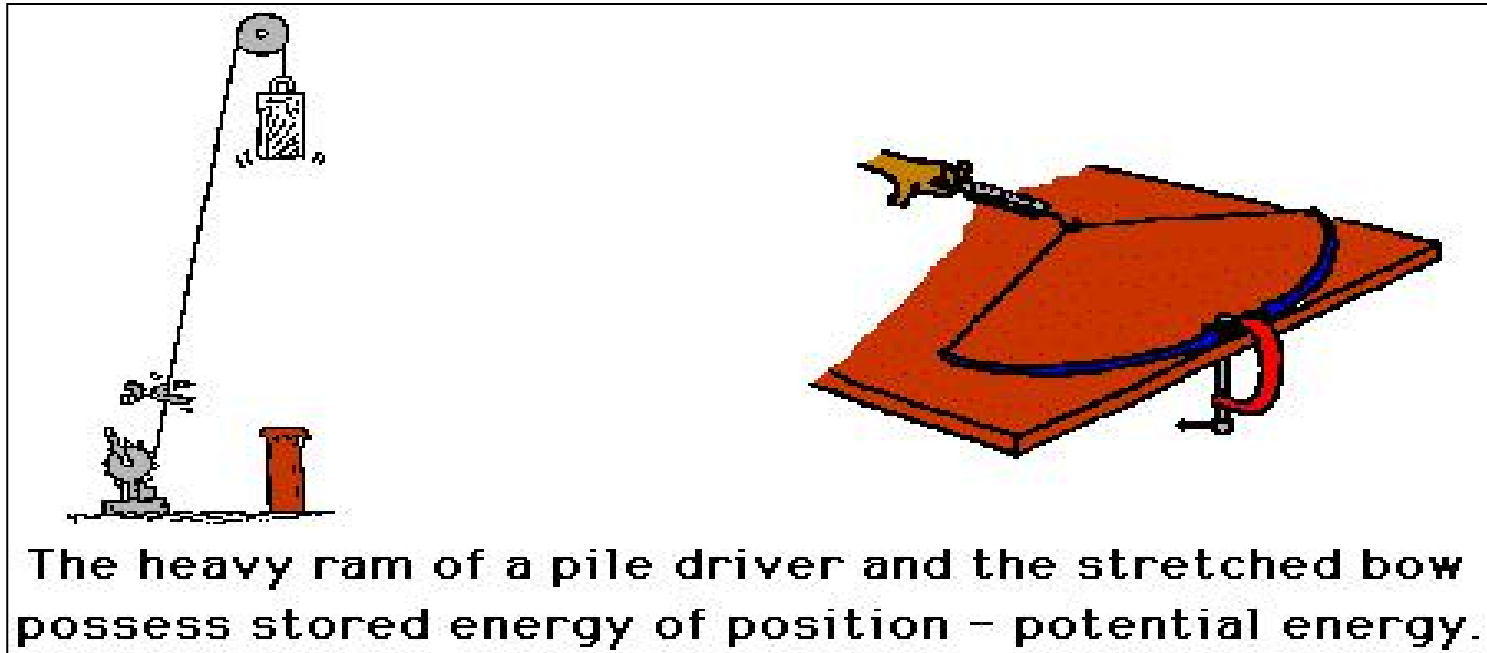
Potential Energy

Potential energy is the energy stored in an object due to its position, state or shape.

In mechanics,

i) Gravitational Potential Energy, U_g

ii) Elastic Potential Energy, U_s



The heavy ram of a pile driver and the stretched bow possess stored energy of position – potential energy.

For example, the heavy ram of a pile driver is **storing energy when it is held at an elevated position.**

Similarly, a drawn bow is able to **store energy** as the result of its **position**. When assuming its *usual position* (i.e., when not drawn), there is no energy stored in the bow. Yet when its position is altered from its usual equilibrium position, the bow is able to store energy by virtue of its position. **This stored energy of position is referred to as potential energy.**

Gravitational Potential Energy, U

It is the energy stored in an object **due to its vertical position**.

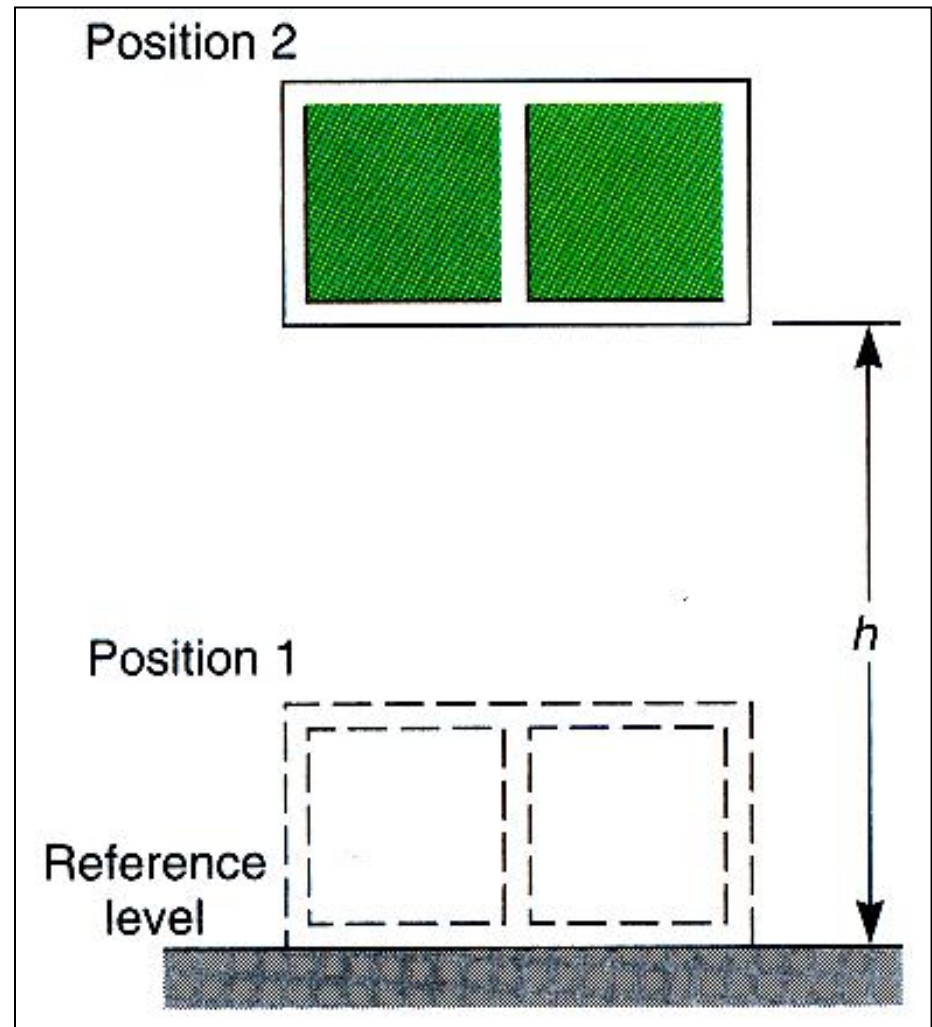
(i.e height)

$$U = mgh$$

m – mass

g – acceleration due to gravity
(9.81 m s^{-2})

h – height



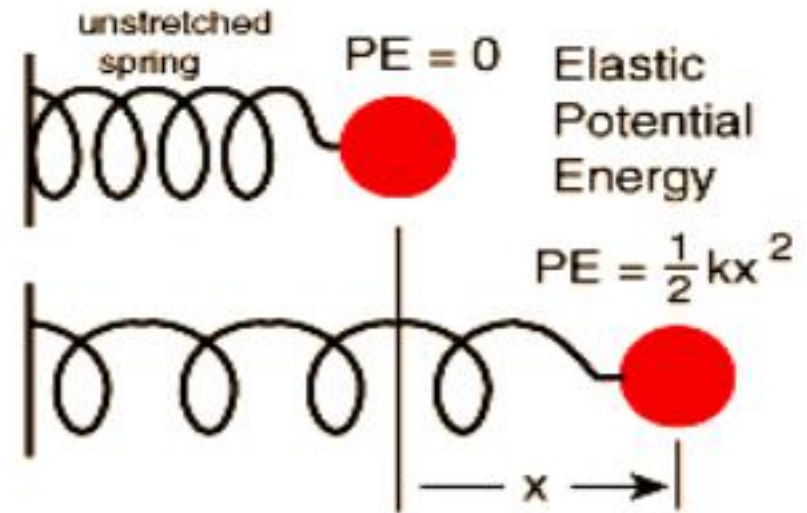
$$U = mgh$$

Gravitational potential energy depends on the mass and the height.



Elastic potential energy, U_s

- Energy stored in elastic materials as the result of their **stretching or compressing**.
- The more stretch, the more stored energy.
- Springs are a special instance of device which can store elastic potential energy due to either compression or stretching.



A force required to stretch a spring.

If the elastic limit is not exceeded, the force applied, F is directly proportional to the amount of stretch or compression, x .

The spring obeys Hooke's law.

$$F \propto x$$

$$\mapsto F = -kx$$

* negative sign in equation indicates that the direction of F is always opposite to the direction of the amount of stretch or compression

The work done to stretch the spring from its equilibrium position to a position, x

$$W = \int_0^x F dx$$

$$W = -k \int_0^x x dx$$

$$W = -\frac{1}{2} kx^2$$

* -ve means work is done to stretch the spring.

The work done causes energy to be transferred and stored in the spring - object system which is known as **elastic potential energy, U_s**

$$U = \frac{1}{2} kx^2$$

where k = spring constant or force constant

x = amount of compression / extension



The compressed springs of a dart gun store elastic potential energy. When the trigger is pulled, the springs apply a force to do work on the dart.

From the Hooke's law (**without “-” sign**), a restoring force, F_s against extension of the spring, x graph is shown in Figure 5.2.

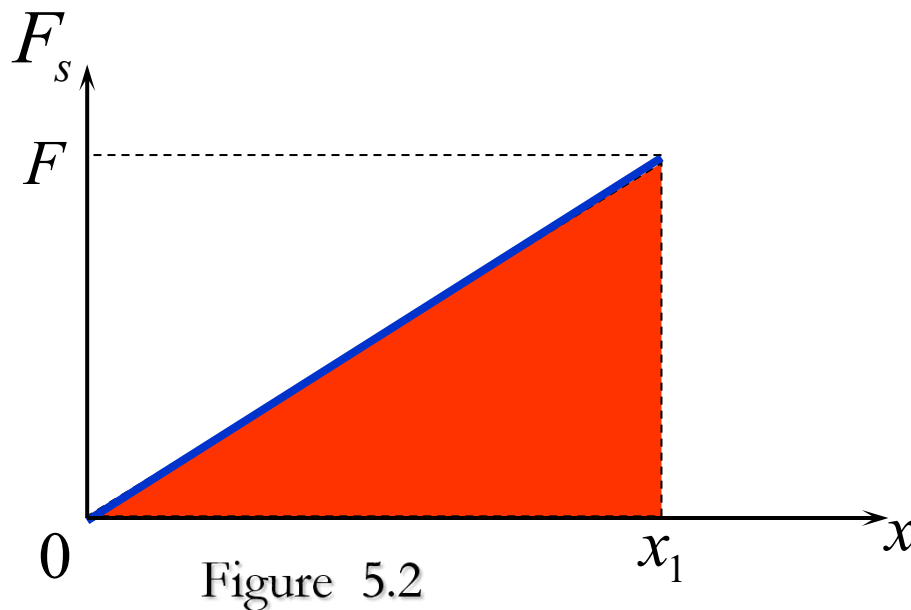


Figure 5.2

$$W = \frac{1}{2} F x_1$$

$$W = \frac{1}{2} (k x_1) x_1$$

$$W = \frac{1}{2} k x_1^2 = U_s$$

→ $W = \text{area under the } F_s - x \text{ graph}$

Kinetic Energy, K

Kinetic energy, K is the energy an object has because of its motion.

Kinetic energy, K of an object of mass m moving with velocity v is defined as

$$K = \frac{1}{2}mv^2$$

-- **Scalar** quantity.

-- Unit SI : **joule (J)**

Conservation of Energy

For an isolated system, the total amount of energy is always constant.

$$\Sigma E_{\text{initial}} = \Sigma E_{\text{final}}$$

The total amount of energy always stays the same.

Energy can be transferred from one form to another but it cannot be destroyed or created.

Conservation of Mechanical Energy

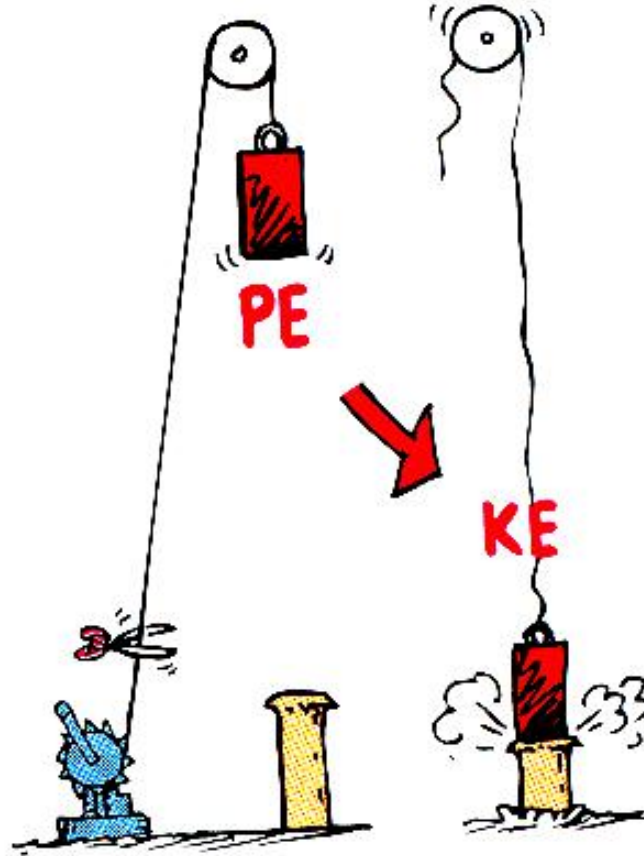
In a conservative system (for example no friction), the total mechanical energy (Kinetic + Potential) is constant.

$$K_{initial} + U_{initial} = K_{final} + U_{final}$$

If the system has friction, then we have:

$$K_{initial} + U_{initial} = K_{final} + U_{final} + W_{fiction}$$

Example

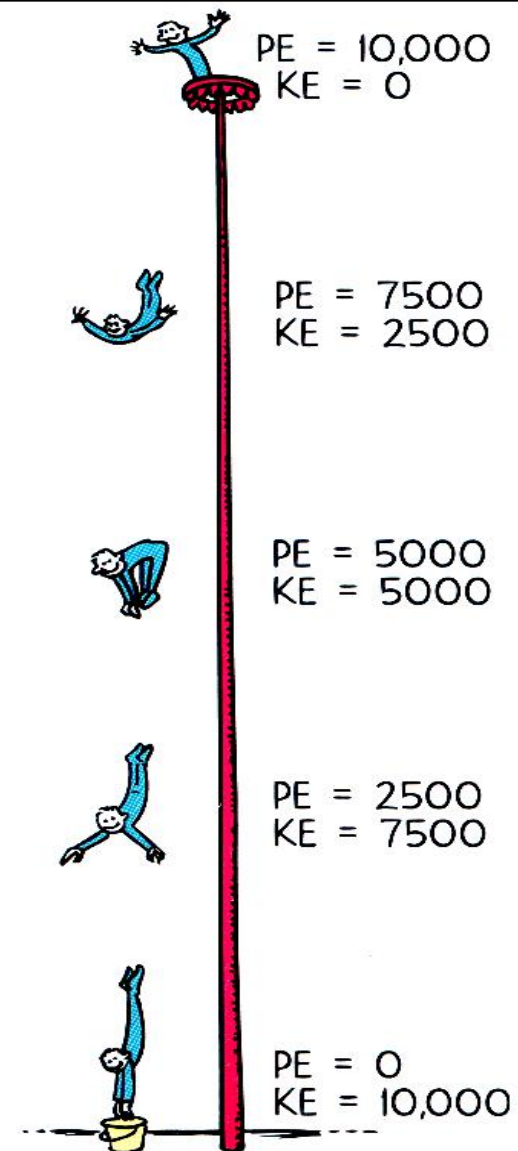


The Potential energy of the elevated ram is converted to kinetic energy when the ram is released.

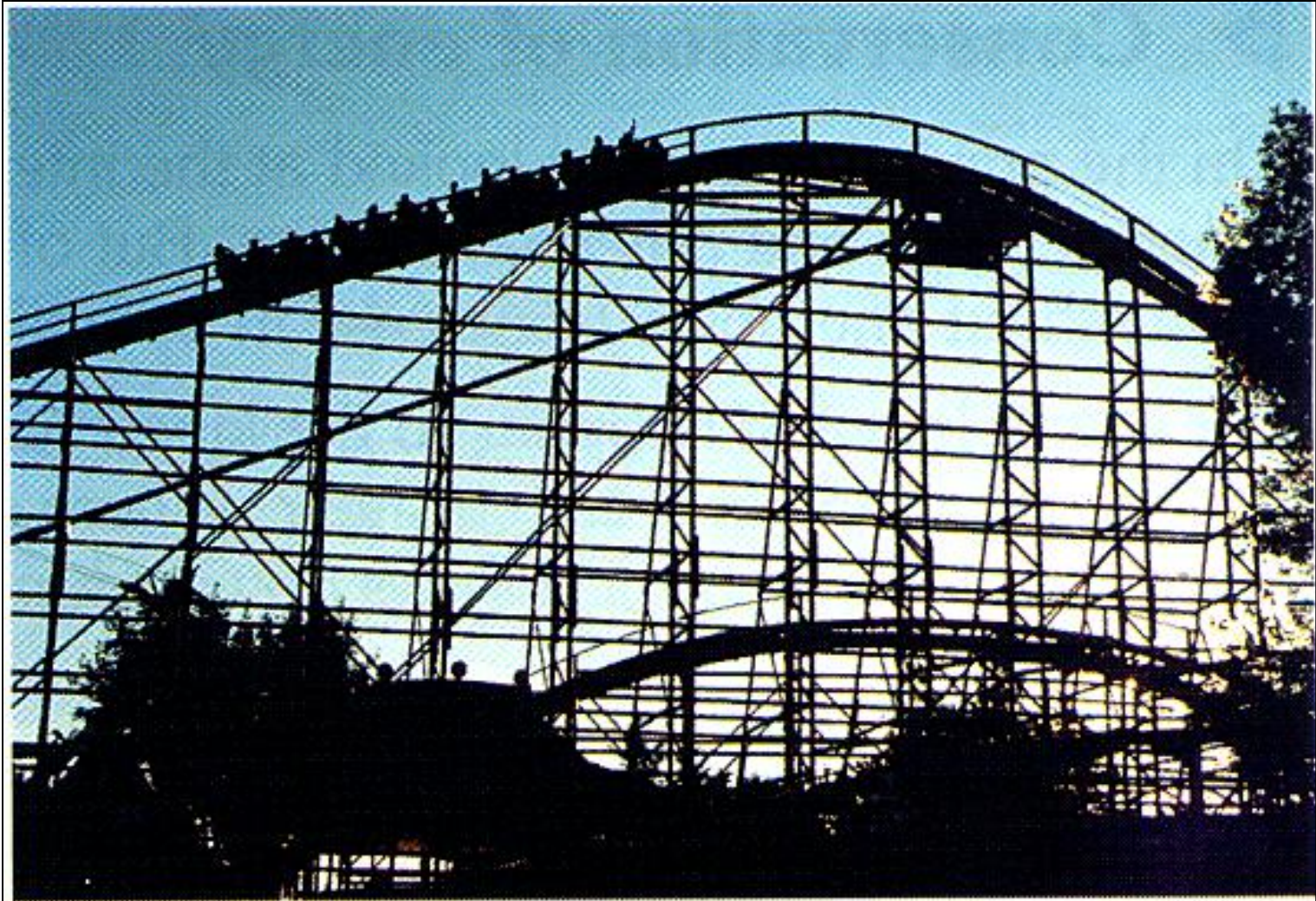
Example

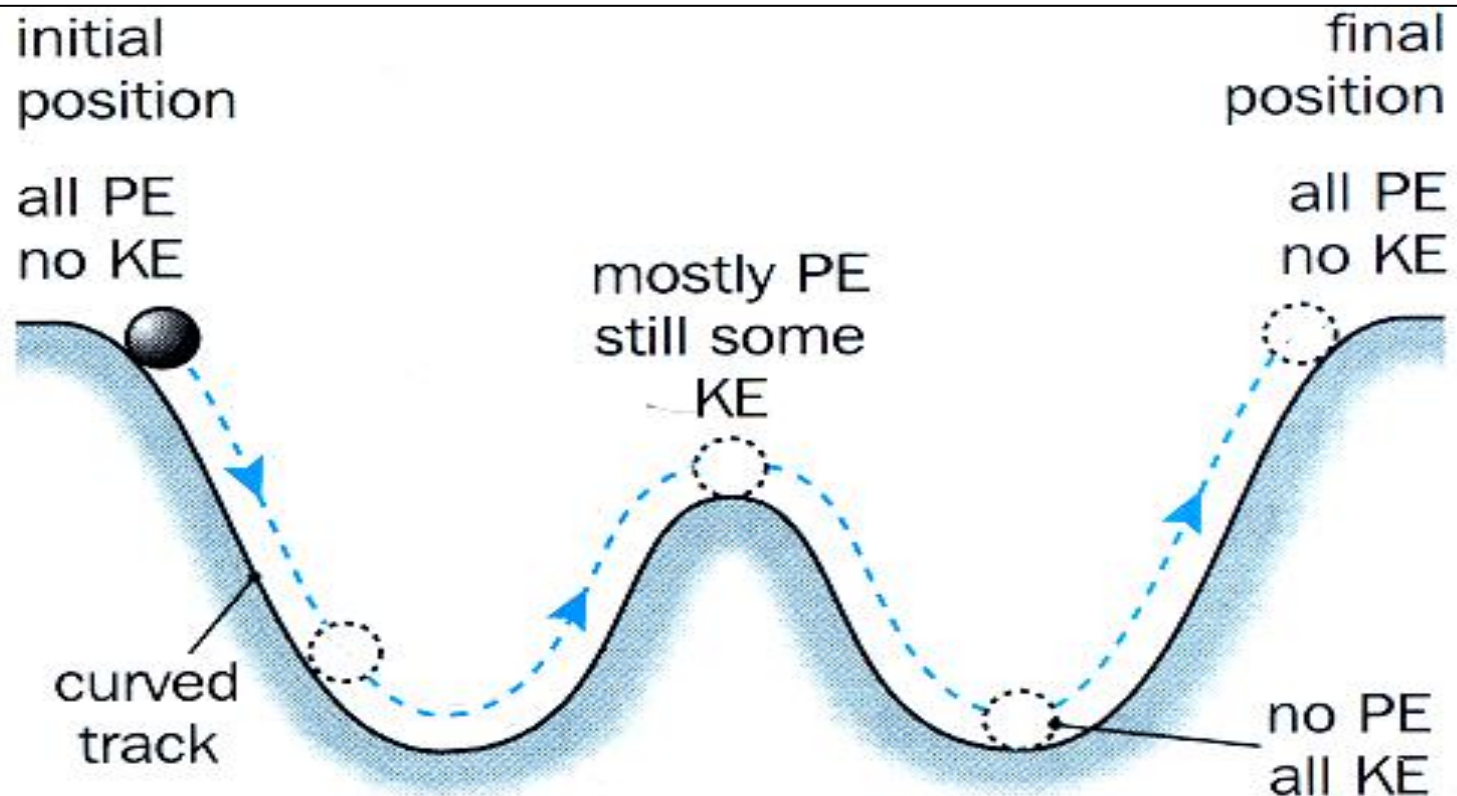
A circus diver at the top of a pole has a potential energy of 10 000 J. As he dives, his potential energy transforms to kinetic energy.

All the way down, the total energy is constant.



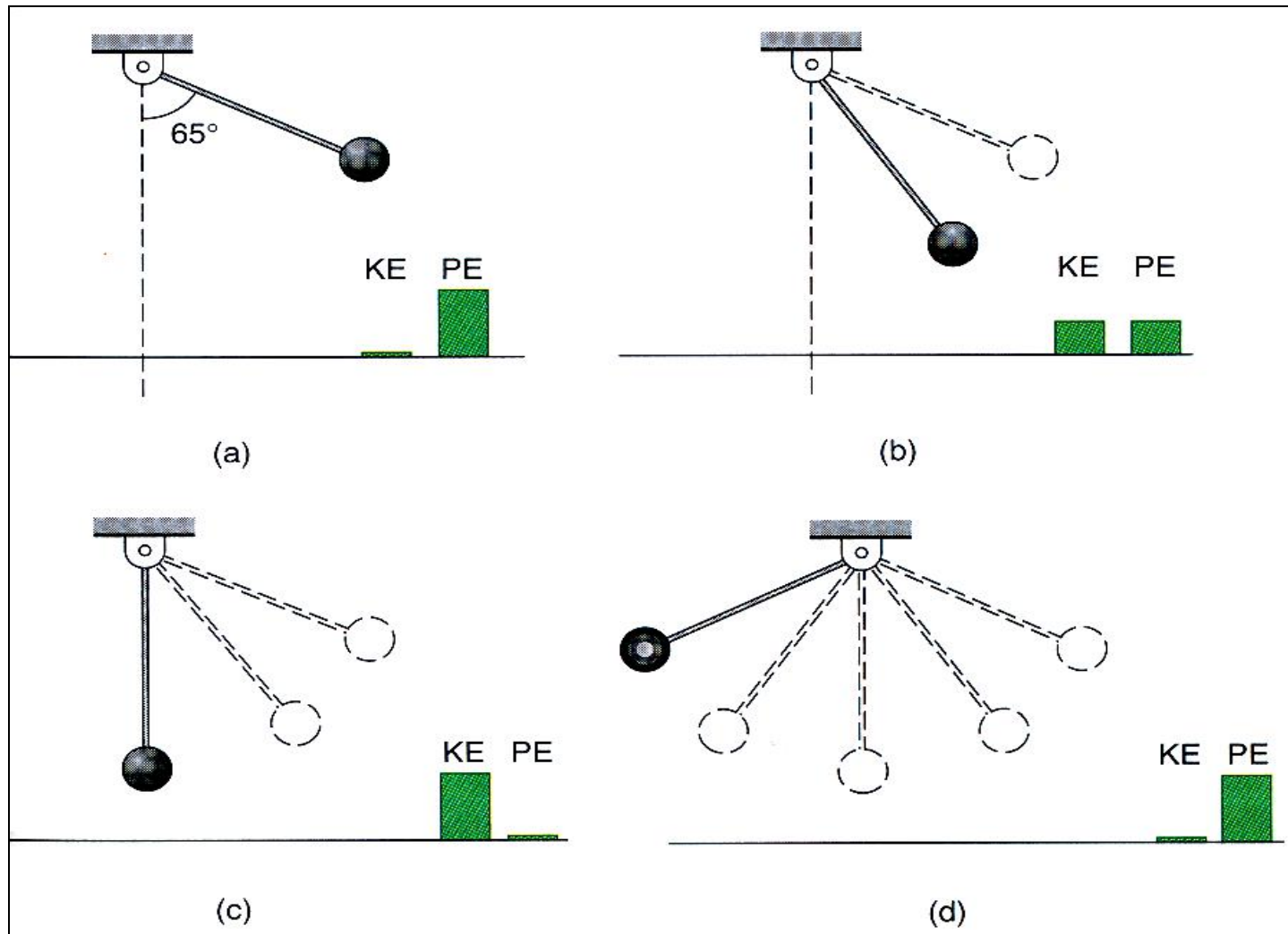
Example : Riding Roller Coaster





With no friction, the ball will roll until it reaches its initial height, regardless of the shape of the track

U and K changes in a swinging pendulum



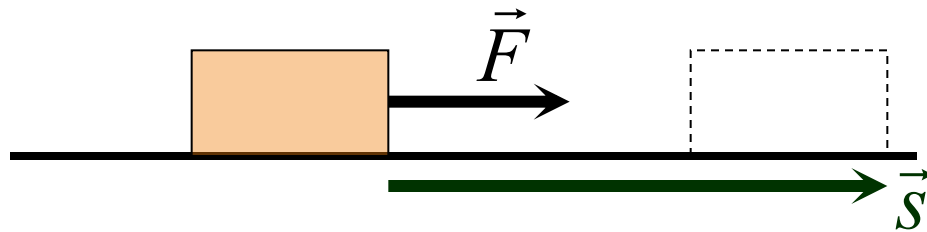


End of Chapter 5

Q1

LO 5.1 c

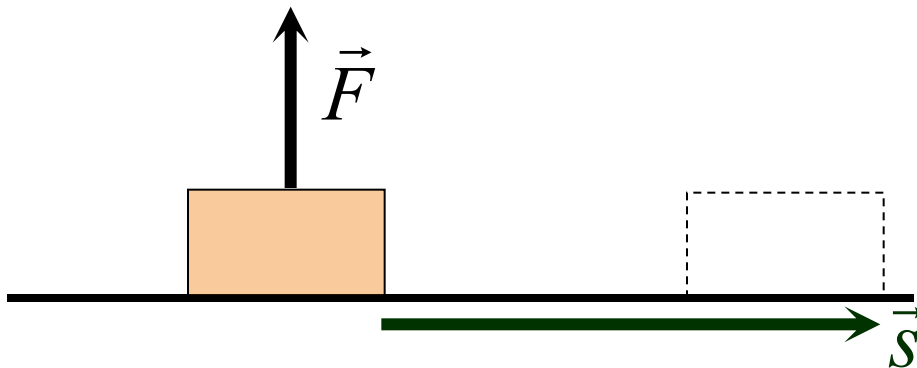
a) Work done by a horizontal force, F on an object.



$$W = Fs \cos \theta \text{ and } \theta = 0^\circ$$

$$W = Fs$$

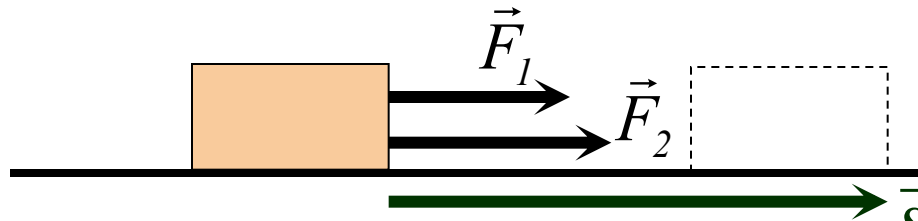
b) Work done by a vertical force, F on an object.



$$W = Fs \cos \theta \text{ and } \theta = 90^\circ$$

$$W = 0 \text{ J}$$

c) Work done by a horizontal forces, F_1 and F_2 on an object.



$$W_1 = F_1 s \cos 0^\circ$$

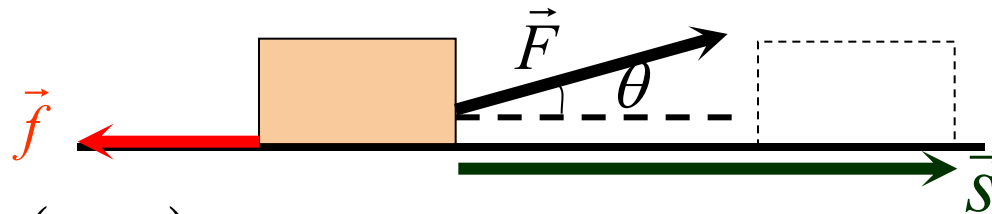
$$W_2 = F_2 s \cos 0^\circ$$

$$\sum W = W_1 + W_2 = (F_1 s + F_2 s)$$

$$\sum W = (F_1 + F_2) s \text{ and } F_{\text{nett}} = F_1 + F_2$$

$$\sum W = W_{\text{nett}} = (F_{\text{nett}}) s$$

d) Work done by a force, F and frictional force, f on an object.



$$W_{\text{nett}} = (F_{\text{nett}}) s \text{ and } F_{\text{nett}} = F \cos \theta - f = ma$$

$$W_{\text{nett}} = (F \cos \theta - f) s \text{ OR } W_{\text{nett}} = mas$$

Q2

LO 5.1 c

You push your physics reference book 1.50 m along a horizontal table with a horizontal force of 5.00 N. The frictional force is 1.60 N.

Calculate

- the work done by the 5.00 N force,
- the work done by the frictional force,
- the total work done on the book.

Solution:

given : $s = 1.50 \text{ m}$, $F = 5.00 \text{ N}$, $f = 1.60 \text{ N}$

a. the work done by the 5.00 N force,

$$\begin{aligned} W_F &= F.s \cos \theta \quad \text{and } \theta = 0^\circ \\ &= (5.00)(1.50) \cos 0^\circ \end{aligned}$$

$$W_F = 7.50 \text{ J}$$

b. the work done by the frictional force,

$$\begin{aligned} W_f &= f.s \cos \theta \quad \text{and } \theta = 180^\circ \\ &= (1.60)(1.50) \cos 180^\circ \end{aligned}$$

$$W_f = -2.40 \text{ J}$$

c. the total work done on the book.

$$\begin{aligned}\sum W &= W_F + W_f \\ &= 7.50 + (-2.40) \\ \sum W &= 5.10 \text{ J}\end{aligned}$$

OR

$$\sum W = F_{\text{net}} s$$

$$\sum W = (F - f)s$$

$$\sum W = (5.00 - 1.60)(1.50)$$

$$\sum W = 5.10 \text{ J}$$

Q3

LO 5.1 c

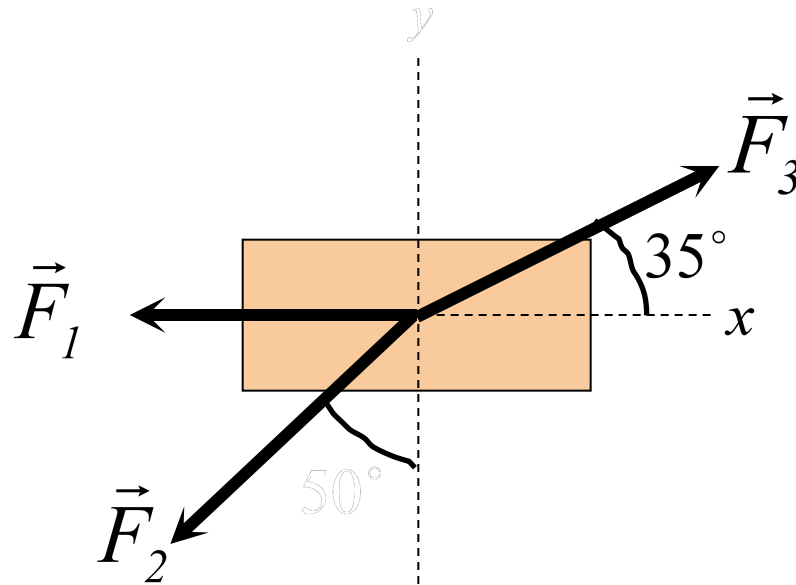


Figure above shows an overhead view of three horizontal forces acting on a cargo that was initially stationary but that now moves across a frictionless floor horizontally. The force magnitudes are $F_1 = 3.00 \text{ N}$, $F_2 = 4.00 \text{ N}$ and $F_3 = 10.0 \text{ N}$. Determine the total work done on the cargo by the three forces during the first 4.00 m of displacement.

Solution:

Given $F_1 = 3 \text{ N}$, $F_2 = 4 \text{ N}$, $F_3 = 10 \text{ N}$, $s = 4 \text{ m}$

$$W = F.s \cos \theta$$

$$W_1 = (3)(4) \cos 180 = - 12.00 \text{ J}$$

$$W_2 = (4)(4) \cos 140 = - 12.26 \text{ J}$$

$$W_3 = (10)(4) \cos 35 = 32.77 \text{ J}$$

$$W_T = W_1 + W_2 + W_3$$

$$= - 12.00 + (- 12.26) + 32.77 = \underline{8.51 \text{ J}}$$

Q4

LO 5.1 c

A block of mass 2.50 kg is pushed 2.20 m along a frictionless horizontal table by a constant 16.0 N force directed 25.0° below the horizontal. Determine the work done on the block by

- the applied force,
- the normal force exerted by the table, and
- the gravitational force.
- Determine the total work on the block.

(Given $g = 9.81 \text{ m s}^{-2}$)

Solution:

Given, $m = 2.5 \text{ kg}$, $s = 2.2 \text{ m}$, $F = 16.0 \text{ N}$, $\theta = 25.0^\circ$

$$\text{a) } W_F = F \cdot s \cos \theta = (16)(2.2) \cos 25 = \underline{31.9 \text{ J}}$$

$$\text{b) } W_N = \underline{0 \text{ J}}, \theta = 90^\circ$$

$$\text{c) } W_g = \underline{0 \text{ J}}, \theta = 90^\circ$$

$$\begin{aligned} \text{d) } W_T &= W_F + W_N + W_g \\ &= 31.9 + 0 + 0 = \underline{31.9 \text{ J}} \end{aligned}$$

Q5

LO 5.1 d

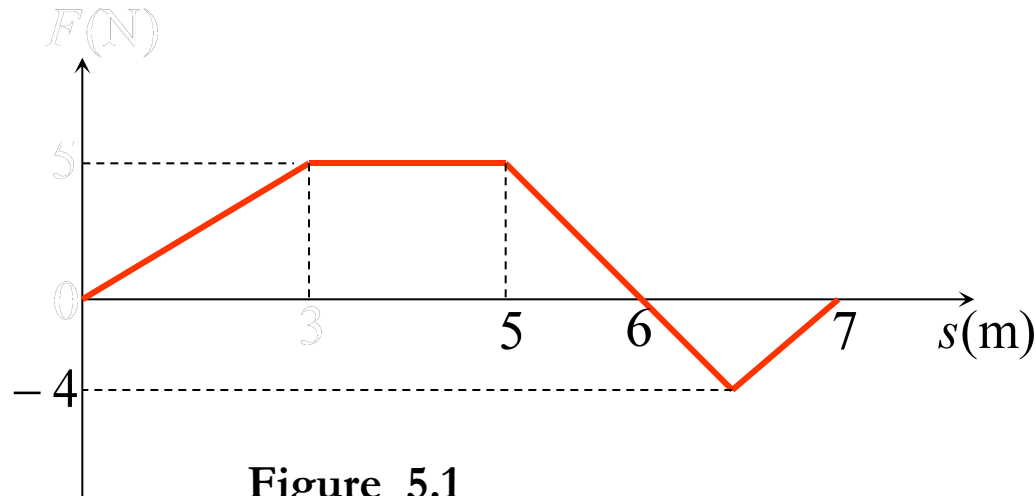


Figure 5.1

A horizontal force F is applied to a 2.0 kg radio-controlled car as it moves along a straight track. The force varies with the displacement of the car as shown in Figure 5.1. Calculate the work done by the force F when the car moves from 0 to 7 m.

Solution:

W = area under the F - s graph

$$= \frac{1}{2}(6 + (5 - 3))5 + \frac{1}{2}(7 - 6)(-4)$$

$$W = 18 \text{ J}$$

Q6

LO 5.2 d

A force of magnitude 800 N caused an extension of 20 cm on a spring. Determine the **elastic potential energy** of the spring when

- a. the extension of the spring is 30 cm.
- b. a mass of 60 kg is suspended vertically from the spring.

(Given $g = 9.81 \text{ m s}^{-2}$)

Solution:

given: $F = 800 \text{ N}; x = 0.200 \text{ m}$

a. the extension of the spring is 30 cm.

From the Hooke's law,

$$F = kx$$

$$800 = k(0.20)$$

$$k = 4 \times 10^3 \text{ N m}^{-1}$$

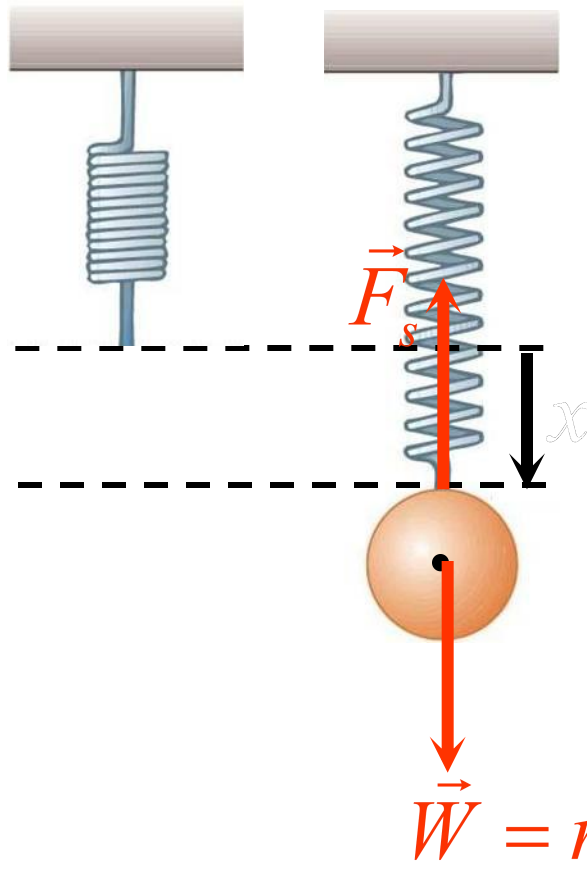
The elastic potential energy

$$U_s = \frac{1}{2} kx^2$$

$$U_s = \frac{1}{2} (4 \times 10^3) (0.300)^2$$

$$U_s = 180 \text{ J}$$

b. Given $m=60$ kg. When the spring is in equilibrium, thus



$$F_{net} = 0$$

$$F_s = mg$$

$$kx = mg$$

$$(4 \times 10^3)x = (60)(9.81)$$

$$x = 0.147 \text{ m}$$

Therefore

$$U_s = \frac{1}{2}kx^2$$

$$= \frac{1}{2}(4 \times 10^3)(0.147)^2$$

$$U_s = 43.2 \text{ J}$$

Q7

LO 5.2 d

In a smooth pulley system, a force F is required to bring an object of mass 5.00 kg to the height of 20.0 m at a constant speed of 3.00 m s^{-1} as shown in Figure 5.2. Determine

- the force, F
- the work done by the force, F .

(Given $g = 9.81\text{ m s}^{-2}$)

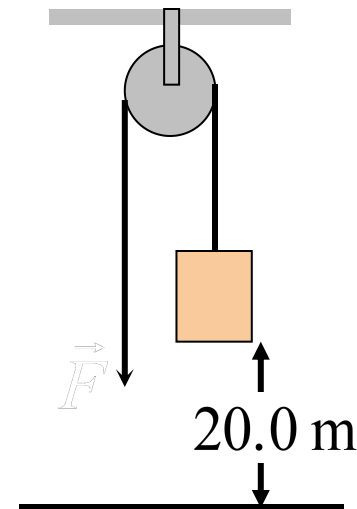


Figure 5.2

Solution:

Given, $m = 5 \text{ kg}$, $h = 20.0 \text{ m}$, $u = 3.0 \text{ ms}^{-1}$

a) $F = ma$

Where $a = g$

$$F = (5)(9.81) = \underline{49.05 \text{ N}}$$

b) $W = F.s \cos \theta$

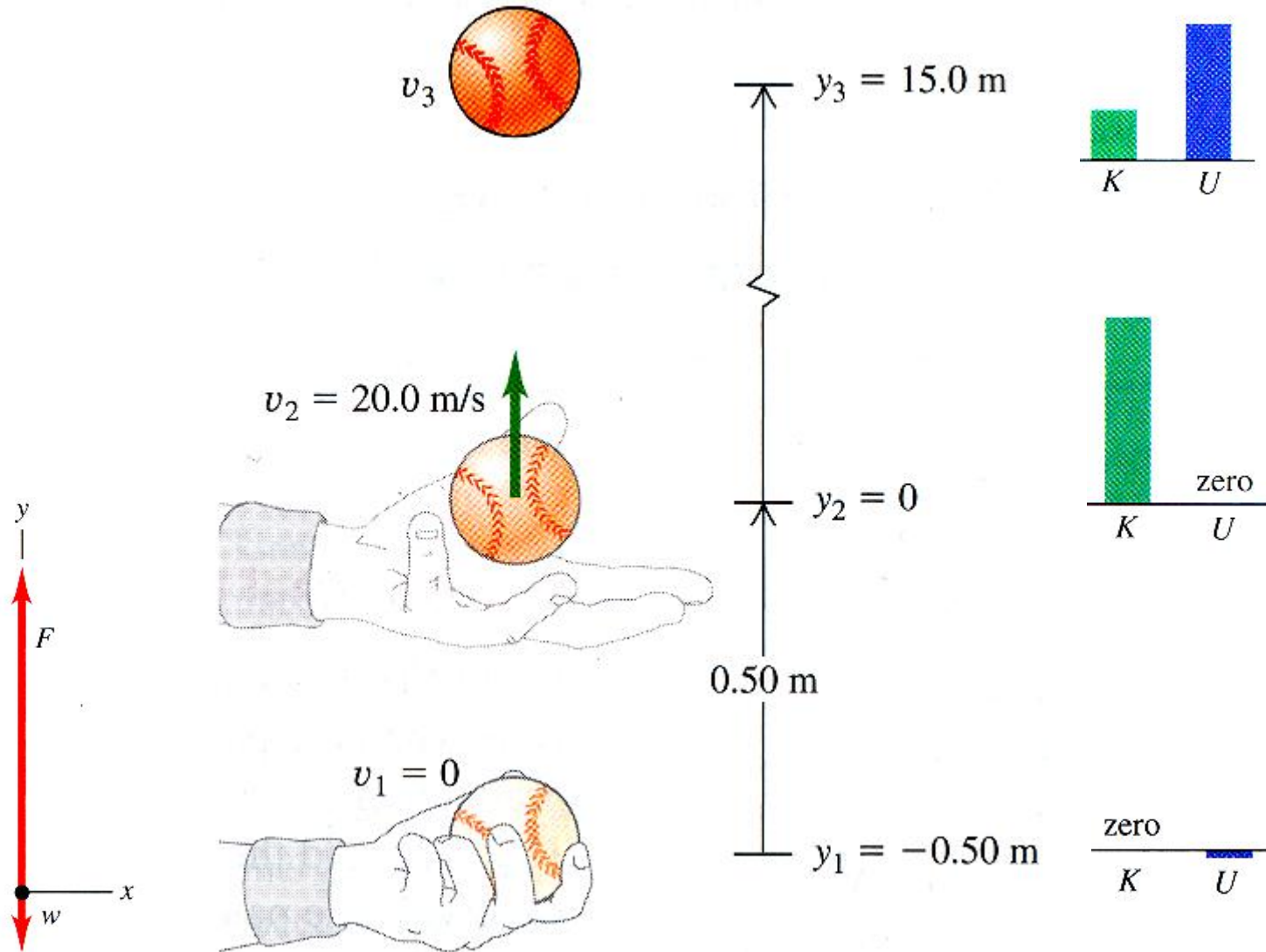
$$= (49.05)(20) \cos 0 = \underline{981 \text{ J}}$$

Q8

LO 5.2 d

Suppose your hand moves up 0.50 m while you are throwing the ball, which leaves your hand with an upward velocity of 20.0 m s^{-1} . Again ignore air resistance.

- (a) Assuming that your hand exerts a constant upward force on the ball, find the magnitude of that force.
- (b) Find the speed of the ball at a point 15.0 m above the point where it leaves your hand.



Solution:

(a) Apply Conservation of energy

$$K_i + U_i + W = K_f + U_f$$

$$0 + mgy_1 + W = \frac{1}{2}mv_2^2 + 0$$

$$(0.145)(9.81)(-0.5) + W = \frac{1}{2}(0.145)(20)^2$$

$$W = 29.71\text{J}$$

Assuming that the upward force is F :

$$W = Fs$$

$$F = \frac{W}{s} = \frac{29.71}{0.5}$$

$$F = 59.42 \text{ N}$$

(b) Assume velocity at 15 m is v_3

Apply Conservation of energy

$$K_i + U_i = K_f + U_f$$

$$K_2 + U_2 = K_3 + U_3$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_3^2 + mgh$$

$$\frac{1}{2}(0.145)(20)^2 = \frac{1}{2}(0.145)v_3^2 + (0.145)(9.81)(15)$$

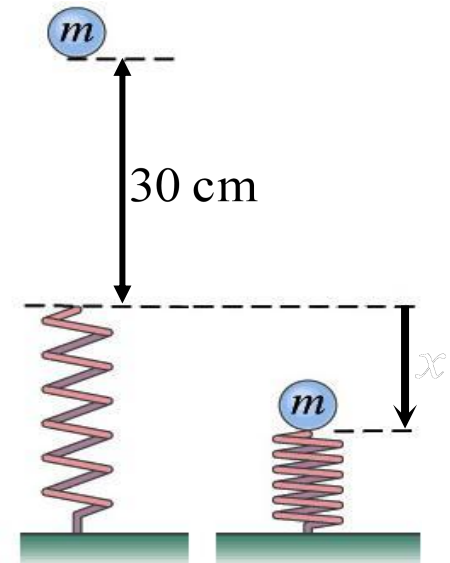
$$v_3 = 10.28 \text{ m s}^{-1}$$

Q9

LO 5.2 d

A 1.5 kg sphere is dropped from a height of 30 cm onto a spring of spring constant, $k = 2000 \text{ N m}^{-1}$. After the block hits the spring, the spring experiences maximum compression, x as shown in Figure 5.3.

- Calculate the speed of the sphere just before strikes the spring.
- Determine the maximum compression, x .
(Given $g = 9.81 \text{ m s}^{-2}$)



Before

After

e Figure 5.3

Solution:

Given, $m = 1.4 \text{ kg}$, $k = 2000 \text{ N m}^{-1}$, $s = x$

a) $E_f = E_i$

$$\frac{1}{2}mv^2 = mgh$$

$$v^2 = (2)(9.81)(0.3) = 5.886$$

$$v = \underline{2.43 \text{ m s}^{-1}}$$

b) $E_f = E_i$

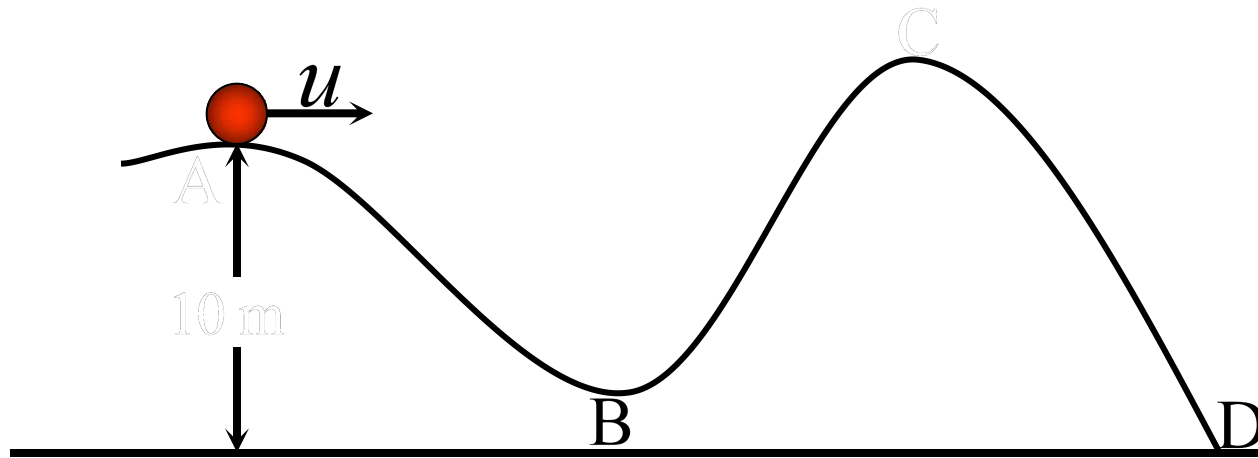
$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$x^2 = \frac{(1.5)(2.43^2)}{2000}$$

$$x^2 = 4.43 \times 10^{-3}$$

Q10

LO 5.2 d



A ball of mass 0.50 kg is at point A with initial speed, $u = 4 \text{ m s}^{-1}$ at a height of 10 m as shown in figure (Ignore the frictional force).

Determine

- the total energy at point A,
- the speed of the ball at point B where the height is 3 m,
- the speed of the ball at point D,
- the maximum height of point C so that the ball can pass over it.

Solution:

Given, $m = 0.5 \text{ kg}$, $h = 10 \text{ m}$, $u = 4 \text{ m s}^{-1}$

$$\begin{aligned} a) \quad E_A &= mgh_a + \frac{1}{2}mv_a^2 \\ &= (0.5)(9.81)(10) + \frac{1}{2}(0.5)(4^2) \\ &= \underline{53.05 \text{ J}} \end{aligned}$$

$$\begin{aligned} b) \quad \sum E_i &= \sum E_f \\ mgh_a + \frac{1}{2}mv_a^2 &= mgh_b + \frac{1}{2}mv_b^2 \\ 53.05 &= (0.5)(9.81)(3) + \frac{1}{2}(0.5)v_b^2 \\ v_b &= \underline{12.38 \text{ m s}^{-1}} \end{aligned}$$

$$c) \sum E_i = \sum E_f$$

$$mgh_d + \frac{1}{2}mv_d^2 = mgh_b + \frac{1}{2}mv_b^2$$

$$0 + \frac{1}{2}(0.5)v_d^2 = (0.5)(9.81)(3) + \frac{1}{2}(0.5)(153.32)$$

$$v_d^2 = 212.18$$

$$v_d = \underline{14.57 \text{ m s}^{-1}}$$

$$d) \text{ h max, } v = 0 \text{ m s}^{-1}$$

$$\sum E_i = \sum E_f$$

$$mgh_c = \frac{1}{2}mv_d^2$$

$$h_c = \frac{14.57^2}{2 \times 9.81} = 10.81 \text{ m}$$